Two day testing and speed plan

Sunday 25 October and Monday 26 October 2026 | All four workspaces and shared services

Purpose

Verify Hynox Campus feature by feature, review the agent-written code, fix confirmed defects, and measure the reported lag before and after changes. The checklist includes 41 module groups and shared cross-cutting checks.

WHAT THE DIFFERENT TESTS MEAN

U Unit

A small function or component in isolation: for example, scoring a quiz answer or rejecting an oversized resume field. Dependencies are mocked where appropriate.

I Integration

Connected parts together: a server action, authentication, database constraints and policies, or an email/worker contract. Mocks alone do not prove database permissions.

E End to end

A real user journey through the browser, such as teacher assignment to student submission and review.

S Security

Try the wrong role, wrong owner, wrong campus, expired session and direct request. Hidden buttons do not prove access is denied.

A Accessibility

Keyboard navigation, labels, focus, announcements, contrast and narrow-screen use.

P Performance

Measure waiting, network requests, rendering, queries and concurrent work. Identify the bottleneck before changing code.

C and R

C = read and review relevant code. R = rerun the failed case and nearby workflows after the fix.

HOW TO USE THE CHECKBOXES

Mark a module complete only after code review, its applicable tests, the failure-path checks, any required fix and a regression rerun. A blocked or failed module stays unchecked and gets an issue ID in the evidence log. These are completion boxes, not promises that every test will pass.

☐ Confirm the two-day working window, tester and reviewer, and a separate test environment.

Scope is comprehensive at feature-group level. New paths found during review must be added to the ledger; two days cannot guarantee every input combination or every defect is resolved.

Day by day execution

Assumption: 8 working hours per day, excluding breaks. Rebalance within the same priorities when evidence changes.

DAY 1 - SUNDAY 25 OCTOBER

- [ ] 1.0 h - Prepare fixtures and isolation; capture commit, environment and existing test/type/lint/build failures.
- [ ] 1.5 h - Review authentication, direct-request authorization, tenant boundaries, RLS and sensitive data handling across all modules.
- [ ] 3.0 h - Execute modules 01-20 on pages 3-5: account access, onboarding, library, academics and delivery.
- [ ] 1.0 h - Fix the highest-impact confirmed defects in shared code; add a small regression test for each non-trivial fix.
- [ ] 0.5 h - Record performance baseline for all dashboards and the slowest journeys using production-mode builds.
- [ ] 1.0 h - Rerun affected checks; record failed/blocked cases and agree Day 2 priorities.

DAY 2 - MONDAY 26 OCTOBER

- [ ] 3.0 h - Execute modules 21-41 on pages 6-8: progress, assessments, attendance, analytics, career and newsletters.
- [ ] 1.5 h - Fix and retest critical/high defects; include data integrity, duplicate submissions and worker recovery.
- [ ] 2.0 h - Profile the worst measured delays, apply the smallest useful fixes and repeat the same measurements.
- [ ] 1.0 h - Run role-specific end-to-end journeys, accessibility smoke checks and the final automated regression.
- [ ] 0.5 h - Finish the evidence ledger and pass/fail/blocked summary; decide whether the build is ready or needs more time.

PRIORITY AND TIME LIMITS

P0: unauthorized access, exposed secrets, data loss or unsafe execution. P1: login failure, broken core workflow, wrong grades or duplicate mail. P2: serious lag, broken recovery or accessibility. P3: minor visual issues. Record expected impact and reproduction; do not use priority labels as proof of exploitability.

If P0/P1 work exceeds the time budget, stop optional polishing and preserve incomplete checks. Do not deploy with unresolved critical security or data-integrity defects. Extend the plan with explicit owners and dates rather than silently reducing coverage.

Day 1 account and onboarding

Modules 01-07 | For every row: code review C, applicable tests below, then fix and regression R.

- [ ] 01 Login and sign out

  Tests: U I E S | Inputs: Valid/invalid password; Google callback success/error; expired session; missing business user; logout then Back.

  Pass: Correct role landing; safe callback destination; protected routes deny stale/anonymous sessions.

- [ ] 02 Public signup and invitation verification

  Tests: U I E S | Inputs: Missing/malformed fields; duplicate email; invalid, expired, reused or wrong-email token; interrupted acceptance.

  Pass: Useful errors; no extra account/role; acceptance is consistent and cannot be replayed.

- [ ] 03 Shared dashboard chooser

  Tests: U I E S | Inputs: Mapped/unmapped role; tampered role cookie/header; inactive account; refresh, deep link and sign out.

  Pass: Only configured roles are selectable; client-supplied context does not grant extra privileges.

- [ ] 04 Institution directory and profiles

  Tests: U I E S | Inputs: Valid/duplicate code; blank/long profile; status transition; invalid ID; delete with dependent records.

  Pass: Correct validation and confirmation; no orphaned records; non-Admin/direct requests denied.

- [ ] 05 Institution user directory and removal

  Tests: I E S A | Inputs: Search/filter empty and large roster; wrong campus ID; remove linked user; cancellation and retry.

  Pass: Correct scoping; safe removal; clear empty/error state; no accidental unrelated deletion.

- [ ] 06 Single user onboarding

  Tests: U I E S | Inputs: Each supported role; conflicting email role; missing institution; Institution Admin inviting staff or other campus.

  Pass: Only allowed role/campus accepted; duplicate retry is safe; invitation mail failure is visible.

- [ ] 07 CSV onboarding

  Tests: U I E S | Inputs: Empty/header-only; quoted commas; BOM/CRLF; malformed/oversized data; duplicate and mixed invalid rows.

  Pass: Validate whole input before changes; reject unauthorized campus/role; no silent partial onboarding.

Common negative cases: blank/null, wrong types/IDs, length limits, duplicate and concurrent requests, denied role/tenant/owner, failure/retry. Apply where relevant; record N/A with a reason.

Day 1 administration and academics

Modules 08-14 | For every row: code review C, applicable tests below, then fix and regression R.

- [ ] 08 Invitations and regeneration

  Tests: U I E S | Inputs: Tenant/status/date filters; expired/accepted invitation; wrong-tenant ID; repeated regeneration; Brevo failure.

  Pass: Status and token lifecycle correct; old token invalidated as intended; no out-of-scope action.

- [ ] 09 Super Admin onboarding

  Tests: U I E S | Inputs: Admin success; every non-Admin denial; duplicate/replayed invite; invalid fields and send failure.

  Pass: Dedicated flow grants only intended role; no escalation through ordinary CSV or user invitation.

- [ ] 10 Teacher campus mappings

  Tests: I E S | Inputs: Add/remove mapping; duplicate map; invalid teacher/campus; teacher mapped to multiple campuses.

  Pass: No duplicate or orphan relation; teaching context updates and stale cache is invalidated.

- [ ] 11 Reusable course blueprints

  Tests: U I E S | Inputs: Course/module/lesson/resource CRUD; categories/tags; published/draft; order; invalid URL; dependent deletion.

  Pass: Correct hierarchy and publication rules; Admin-only writes; reads match actual published policies.

- [ ] 12 Blueprint instantiation

  Tests: I E S | Inputs: Clone into valid/invalid program; empty blueprint; repeated request; resource propagation; mid-operation failure.

  Pass: Complete hierarchy and correct tenant; no partial clone or unintended duplication; source retained.

- [ ] 13 Programs and batches

  Tests: U I E S | Inputs: Create/edit/status/slug; duplicate and blank inputs; wrong tenant; program removal with courses.

  Pass: Staff action plus database checks agree; correct sort/filter and safe dependent-record handling.

- [ ] 14 Courses and instructor mapping

  Tests: U I E S | Inputs: CRUD; duplicate slug; invalid program; multiple/removed instructors; cross-campus staff IDs.

  Pass: Only valid relations and permitted tenant; instructor list, syllabus and caches remain consistent.

Common negative cases: blank/null, wrong types/IDs, length limits, duplicate and concurrent requests, denied role/tenant/owner, failure/retry. Apply where relevant; record N/A with a reason.

Day 1 content and delivery

Modules 15-20 | For every row: code review C, applicable tests below, then fix and regression R.

- [ ] 15 Syllabus modules and lessons

  Tests: U I E S A | Inputs: Empty/full content; reorder; lesson types and duration; long text; invalid ID; failed save and retry.

  Pass: Stable order and data; staff controls work; Students cannot author through direct requests.

- [ ] 16 Lesson media and supplementary resources

  Tests: U I E S A | Inputs: Markdown/script-like text; unsafe URL schemes; missing video/file; long URL; mobile/keyboard preview.

  Pass: No script execution; safe links and fallback errors; external file access is handled separately.

- [ ] 17 Cohorts and timelines

  Tests: U I E S | Inputs: Dates reversed/equal/missing; inactive cohort; multi-campus context; edit/delete with enrollments.

  Pass: Date validation and lifecycle correct; verify broad staff action guards against intended permissions.

- [ ] 18 Student enrollment and status

  Tests: U I E S | Inputs: Duplicate enrollment; inactive/wrong-campus student; status changes; remove/re-enroll; failed request.

  Pass: No duplicate active relation; correct roster and access; no lost history or tenant leakage.

- [ ] 19 Cohort course assignments

  Tests: I E S | Inputs: Duplicate assignment; wrong program/tenant; future/inactive course; remove while progress exists.

  Pass: Correct course visibility and dates; safe dependency behavior; progress is preserved as intended.

- [ ] 20 Cohort teacher assignments

  Tests: I E S | Inputs: Assign/remove teacher; duplicate; unassigned teacher request; context switch after removal.

  Pass: Correct teaching scope and roster visibility; revoked assignment does not retain stale authorization.

Common negative cases: blank/null, wrong types/IDs, length limits, duplicate and concurrent requests, denied role/tenant/owner, failure/retry. Apply where relevant; record N/A with a reason.

Day 2 progress and assessments

Modules 21-27 | For every row: code review C, applicable tests below, then fix and regression R.

- [ ] 21 Lesson and course progress

  Tests: U I E S | Inputs: Start/complete twice; partial lesson; empty course; activity completion/rejection; concurrent updates.

  Pass: Idempotent progress and correct totals; only intended student records change; completion recalculates.

- [ ] 22 Activity definitions and deployment

  Tests: U I E S | Inputs: Quiz/project/code CRUD; publish/draft; invalid dates/scores; duplicate deployment; expired assignment.

  Pass: Valid subtype and cohort links; availability enforced on server; no half-created activity.

- [ ] 23 Quiz questions options and settings

  Tests: U I E S | Inputs: Question types; zero/multiple correct options; blank question; score limits; reorder/delete; strict-mode toggles.

  Pass: Valid question rules and totals; correct answers are not leaked to an active Student attempt.

- [ ] 24 Quiz attempts answers and scores

  Tests: U I E S | Inputs: Start/resume; unanswered/duplicate answers; exact pass threshold; timeout; double-submit; two tabs.

  Pass: Correct grade, attempts and completion; one consistent final result; no other student access.

- [ ] 25 Quiz retake approval

  Tests: U I E S | Inputs: Approved/denied role; wrong cohort; repeated approval; attempts at limit; previous forfeiture.

  Pass: Retake state and attempt count correct; only intended reviewer can authorize; prior evidence retained.

- [ ] 26 Project definitions and submission

  Tests: U I E S | Inputs: Valid/invalid repository URL; notes length; late/early submission; duplicate submit; rejected/revision resubmit.

  Pass: Only eligible submissions accepted; latest allowed state shown; no overwrite of unrelated history.

- [ ] 27 Project reviews grades and feedback

  Tests: U I E S | Inputs: Approve/reject/revise; score outside limits; empty feedback; duplicate or simultaneous reviews; wrong reviewer.

  Pass: Authorized reviewer only; stable score and feedback; activity/course progress stays consistent.

Common negative cases: blank/null, wrong types/IDs, length limits, duplicate and concurrent requests, denied role/tenant/owner, failure/retry. Apply where relevant; record N/A with a reason.

Day 2 coding attendance and analytics

Modules 28-34 | For every row: code review C, applicable tests below, then fix and regression R.

- [ ] 28 Programming challenges examples and tests

  Tests: U I E S | Inputs: Problem CRUD; zero tests; hidden/visible cases; wrong expected output; limits, languages and max attempts.

  Pass: Only valid configuration accepted; hidden test data stays private; deleting cases preserves consistency.

- [ ] 29 Challenge starter templates

  Tests: U I E S | Inputs: Admin, assigned/unassigned Teacher, Trainer, Institution Admin and Student; every language; availability edges.

  Pass: Read/write rules match approved policy; explicitly resolve teacher-versus-trainer policy difference.

- [ ] 30 Coding execution and submissions

  Tests: U I E S | Inputs: Correct/compile/runtime/timeout/memory errors; empty and large code/stdin; queue retry, worker crash and concurrency.

  Pass: Safe sandbox; no network/files escape; no hidden-test leak; idempotent judging and accurate grade/logs.

- [ ] 31 Strict mode and forfeit endpoints

  Tests: U I E S A | Inputs: Fullscreen/visibility loss; refresh/close; duplicate beacon; forged attempt/student ID; missing auth; completed attempt.

  Pass: Only current owner/active attempt affected; no double penalty; recovery and instructions are accessible.

- [ ] 32 Attendance register and records

  Tests: U I E S | Inputs: Present/absent/late; bulk/individual; invalid student; duplicate save; server midnight; past/future date.

  Pass: Correct totals; Student cannot mark; Teacher date rules and Admin exception enforced server-side.

- [ ] 33 Attendance holidays and statistics

  Tests: U I E S | Inputs: Set/remove holiday; Sunday; no sessions; zero denominator; same-campus unassigned teacher; cross-campus request.

  Pass: Correct holiday/statistics math; verify assignment and same-institution fallback against intended access.

- [ ] 34 Campus analytics and student drilldown

  Tests: U I E S P | Inputs: Empty campus/cohort; multiple groups; incomplete/forfeited work; score bands; roster filters and rank ties.

  Pass: Metrics match source records; no NaN/double counting; correct tenant/owner; filter controls do not alter data.

Common negative cases: blank/null, wrong types/IDs, length limits, duplicate and concurrent requests, denied role/tenant/owner, failure/retry. Apply where relevant; record N/A with a reason.

Day 2 career and newsletters

Modules 35-41 | For every row: code review C, applicable tests below, then fix and regression R.

- [ ] 35 Saved resumes and builder

  Tests: U I E S A | Inputs: Every section/field; blank/long/unsafe content; copy/delete; stale version; offline save/retry; PDF with many entries.

  Pass: Owner-only persistence; conflict preserves draft; one reference layout; selectable PDF has no lost/overlapping text.

- [ ] 36 ATS scanner and saved history

  Tests: U I E S | Inputs: Builder/text/PDF; image-only/malformed/large PDF; 5 MB/10-page edges; optional Role/JD alone and together.

  Pass: Correct extraction/term boundaries; null result for unknown terms; private history; raw files/text not retained.

- [ ] 37 Student newsletter subscription

  Tests: U I E S | Inputs: Opt in/out; shared demo; inactive student; suppressed address; no eligible courses; simultaneous preference change.

  Pass: Own preference only; consent recorded; no forced re-enable or suppression bypass.

- [ ] 38 Newsletter administration and previews

  Tests: U I E S A | Inputs: Limits/hour/reserve boundaries; topic length; disabled course; feed error; preview; dashboard deep link/back/refresh.

  Pass: Admin-only access; Newsletter last and embedded; URL synced; preview never sends mail.

- [ ] 39 Newsletter scheduled delivery and suppression

  Tests: U I S P | Inputs: Stale/duplicate news; 429/feed fallback; no credits; opt-out race; worker overlap; timeout after send; webhook replay.

  Pass: One eligible student/day edition; quota and suppression preserved; ambiguous send reconciled, never blindly retried.

- [ ] 40 Lookup catalogs and runtime choices

  Tests: U I E S | Inputs: Inactive/unknown language/status; missing lookup; duplicate code; unsupported runtime; version label mismatch.

  Pass: Valid choices only; no crash on missing data; catalog labels checked separately from actual sandbox images.

- [ ] 41 Public and Mentor placeholder hub

  Tests: I E S A | Inputs: Public, Mentor and other roles; direct staff route/action; no tenant; unprovisioned/inactive user.

  Pass: Correct limited landing; no assumed mentor teaching permissions; protected functionality stays inaccessible.

Common negative cases: blank/null, wrong types/IDs, length limits, duplicate and concurrent requests, denied role/tenant/owner, failure/retry. Apply where relevant; record N/A with a reason.

Performance and code review

Measure on Day 1; fix and compare on Day 2. Proposed internal targets below are provisional acceptance budgets.

MEASURE THE REPORTED LAG

- [ ] Build and start in production mode; record machine/browser, commit, region, network and data volume. Keep development hot-reload timings separate.
- [ ] Time cold login-to-dashboard, warm tab switches, cohort/course selection, Student Hub search, lesson opening, analytics, builder, scanner and coding results. Record 10 repeat runs per journey and median/p95.
- [ ] Capture browser Network/Performance traces: request count and size, long main-thread tasks, React rerenders, navigation/loading behavior and memory after repeated tab switches.
- [ ] Trace slow server actions and database queries. Count duplicate/N+1 queries, excessive prefetch payloads, missing filters/pagination and expensive joins; use safe staging EXPLAIN analysis.
- [ ] Use synthetic small and realistic datasets: 2 campuses, 3 groups/campus, 50 students/group; then a larger 1,000-student dataset if time allows. Stage concurrency at 1, 10 and 25 virtual users.
- [ ] Inspect query keys and invalidation, 60-second staleness, two-minute dashboard polling, focus/reconnect refetches and retries. Ensure caching never leaks another user or campus.
- [ ] Profile heavy imports, tables/charts, PDF processing and Monaco loading. Add lazy loading, pagination, query/index fixes or render reductions only where traces justify them.

PROVISIONAL BEFORE AND AFTER GATES

Warm tab response <= 500 ms; representative read-action p95 <= 1 s; primary dashboard usable <= 2.5 s under the agreed test setup. Investigate main-thread tasks > 50 ms and interaction delays > 200 ms. External mail/news/code execution gets a separate latency/error budget. These are targets, not measured results.

Accept a speed change only if repeat measurements improve the identified bottleneck without worse errors, correctness, accessibility or authorization. Record any missed target and owner. Never trade tenant/owner checks for speed.

CODE REVIEW APPLIED ACROSS EVERY MODULE

Trace UI -> action -> service -> SQL/RPC/external dependency. Inspect validation, authorization, transaction/partial-write behavior, error propagation, stale closures, concurrent updates, safe logging and secret handling. Search every caller before a shared fix; remove unused/repeated work only within the confirmed scope.

Preparation evidence and final gates

This page is used at the start and end of the two-day run. An unchecked critical gate blocks release readiness.

BEFORE THE FIRST TEST

- [ ] Create an isolated test/staging dataset and recovery snapshot; use fictional contacts and test mail destinations. No destructive production tests or real newsletter broadcasts.
- [ ] Prepare Admin, Institution Admin, Teacher, Trainer, Student, Public/Mentor, shared-demo, inactive, unassigned and second-campus accounts. Test direct APIs as well as UI controls.
- [ ] Baseline: npm test; npx tsc --noEmit; npm run lint; npm run build. Record existing failures separately. Review the 25 existing test files before adding cases; do not assume test count means coverage.
- [ ] Review SQL fixtures in supabase/tests. Run career/newsletter SQL tests only against the isolated database. Inspect scratch scripts before any execution; they may mutate shared data.

FAILURE TO FIX TO PROOF

For each failure record: case/module ID, role and tenant, input, expected/actual result, reproduction steps, severity, screenshot/log/trace, root cause, changed files/commit, automated regression, retest result and owner. Redact secrets and personal data. Keep failed/blocked/N/A distinct.

RELEASE READINESS

- [ ] All 41 module groups have a pass/fail/blocked record and evidence. Core journeys work for each role; authorization differences from the reports are resolved or explicitly tracked.
- [ ] End-to-end: onboard -> assign teacher/course/student -> learn -> quiz/project/code -> review -> progress/analytics/attendance; student resume -> scan -> PDF; subscribed test newsletter -> suppression.
- [ ] Rerun full automated suite and affected SQL/integration tests; types, lint and production build have no unexplained new failures. Exercise failure recovery, double submit and two-tab editing.
- [ ] Keyboard and focus checks; mobile 375 px and desktop; refresh/deep links/back/forward; loading, empty, error and offline/reconnect states across all dashboards.
- [ ] No open P0/P1 security, grading, data-loss or core-flow defects. Performance before/after evidence meets agreed budgets or has an explicit unresolved-item decision.
- [ ] Reviewer signs off summary, remaining risks and deployment/rollback steps. Confirm staged fixes and migrations before any separate production release.

Outcome: Pass ____ Fail ____ Blocked ____ N/A ____ Open issues ____

Tester _____ Reviewer _____ Date _____

Basis: current route/action/service inventory, package scripts, Vitest config and earlier 41-module access audit. Planning only; no application tests, fixes or speed improvements executed in this run.